Geometries from linear spaces and Steiner designs

Human Compression project

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The construction and the results recovered

We study the construction of Leemans [L], which extends an earlier construction of Van Nypelseer. The source is the complete fourteen-page journal edition, *A generalization of a construction due to Van Nypelseer*, Beiträge zur Algebra und Geometrie 46 (2005), 561–574. For a Steiner design the construction is precisely the geometry of Part II of the common paper [G]. For an arbitrary partial linear space its rank-three form is given below. We call a firm, residually connected, flag-transitive geometry satisfying $(\mathrm{IP})_2$ an A-geometry. Automorphisms preserve types unless stated otherwise.

The purpose is to recover the principal classification and its geometric and group data, while correcting false intermediate assertions. The common theorems give all chamber and residue reductions. The remaining work consists of explicit rank-three geometries, field actions, and finite Witt-design stabilizers. In particular, residual connectedness and flag transitivity are separate conditions: the second iteration of a deleted affine line is an A-geometry at $q = 8$, rather than the source's $q = 4$; the subline construction is residually connected for every $q \geq 3$ when the extension degree is two, although its flag-transitive case still has $q = 4$.

The full Hermitian-unital proof is included, using the transport theorem of [G, Part III]. Its conjectural statement in [L, §4.6] was already resolved in the earlier accepted project work; this edition does not claim a second result or a new priority. The rank-six Mathieu geometry, the distinct rank-five residues and their actual stabilizer groups are also included. Identifications with Pasini's named geometries $\Gamma_{23}(4)$ and $\Gamma_8(4)$ remain excluded: the available evidence does not supply object-level maps to those earlier definitions. No isomorphism of geometries is inferred merely from matching group names.

Two outside classification theorems

Two substantial classifications are used at their point of application; they are not consequences of the elementary arguments in [G]. Buekenhout, Dehaye and Leemans [BDL, Theorem 6.2, pp.42–43] classify finite linear spaces with a faithful flag-transitive group whose point and line stabilizers are doubly transitive on their respective incident objects. Their possibilities are:

1. $AG(2, 4)$ with $A\Gamma L(1, 16)$;
2. $PG(n, q)$, $n \geq 2$, with $PSL(n + 1, q) \leq G \leq P\Gamma L(n + 1, q)$;
3. the specified A_7 action on $PG(3, 2)$;
4. $AG(n, q)$, $n \geq 2$, $q \geq 3$, with translations and linear stabilizer containing $SL(n, q)$;
5. the classical unital $U_H(q)$, with $PSU(3, q) \leq G \leq P\Gamma U(3, q)$;
6. circles, with the specified triply transitive point actions.

The last case is absent under the thickness hypothesis of [L, Theorem 2]. Small overlapping cases are not separate families; for example $U_H(2) \cong AG(2, 3)$. This input supplies the classification quoted in that

source theorem. The construction-specific conclusions for the projective and affine cases will be proved directly below.

For the higher-rank classification we use Kantor [K, Theorem 3 and its proof, p.73]. For a Steiner system with $k \geq t+1 \geq 4$ and a t-transitive group, it leaves the affine binary plane designs, projective subline designs, and the five Mathieu designs with parameters

$$(4,5,11), \quad (5,6,12), \quad (3,6,22), \quad (4,7,23), \quad (5,8,24).$$

The affine groups are $2^d : GL(d,2)$ and, when $d = 4$, $2^4 : A_7$. In the subline case the blocks are the projective images of $\mathbb{F}_q \cup \{\infty\}$ in $PG(1,q^e)$, $e \geq 2$; the group contains $PSL(2,q^e)$. We also use the upper bound by $P\Gamma L(2,q^e)$ supplied by the proof of [K], not merely the lower bound reproduced in [L, Theorem 10]. The classical actions in this classification, including their identification with the native Mathieu permutation groups in GAP, remain outside inputs. The actual designs, incidences and subgroup calculations are checked separately below.

Rank-three bases and their iterations

For a partial linear space S, write Γ(S) for Construction3, with objects original points P, original lines L and flags (x,M), and incidence

$$P \sim L \iff P \notin L, \quad P \sim (x,M) \iff P \in M \backslash \{x\}, \quad L \sim (x,M) \iff L \cap M = \{x\}. \tag{B1}$$

A chamber is therefore exactly the ordered configuration

$$P - M - x - L, \qquad P \neq x, \quad M \neq L, \tag{B2}$$

where the three successive incidences are incidences in S. Partial linearity makes P∉L automatic. The source calls these "paths of length four"; (B2) specifies unambiguously the four vertices and three edges.

Linear spaces and the necessary classification

Proposition B.Necessity (L092 Theorem1). Let S be a thick connected partial linear space. If Γ(S) is a residually connected geometry and G≤Aut(S) is chamber-transitive on Γ(S), then G is transitive on the configurations (B2), and S is a linear space, the dual of a linear space, or a (4,3,4)-gon.

Proof. The first assertion is the chamber bijection (B2).

For any point p and any nonincident line N of S, N occurs as an object in the point residue Γ(S)p. It cannot be isolated: that residue is a connected rank-two geometry and has objects of both types. Therefore some flag (x,M) is incident to p and N. This gives a path p-M-x-N of length three in S. Incident point-line pairs already have distance one, so every point-line distance is at most three. It follows that any two points, or any two lines, have distance at most four, by adding an incidence at one endpoint.

There is a proper triangle in S. Fix a point p. Its flag objects in Γ(S)p are indexed by its collinear points x≠p, with the unique joining line M=px. Thickness supplies at least two such points on different lines through p. Connectivity of the residue gives a path between their flag objects. Whenever this path passes through an original line N, its two consecutive flag objects correspond to two distinct points x,y on N, each collinear with p, while p∉N. Thus p,x,y are a proper triangle. Partial linearity rules out an incidence cycle of length four, and the proper triangle gives one of length six. The gonality is exactly three.

Both point- and line-diameter are at most four. If the point-diameter is at most three, all point-point distances, being even, are at most two: every pair of points is collinear, so S is a linear space. If the line-diameter is at most three, the dual assertion holds. In the remaining case both diameters equal four, giving (4,3,4). ■

This proof uses the necessary direction of the source's local-connectivity discussion. It does not need to assume that connectivity of a graph on collinear neighbors alone automatically removes isolated anti-flag objects from a residue.

Proposition B.LocalActions (the needed local-action clauses of L092 Lemma3). If S is a thick partial linear space and G is chamber-transitive on Γ(S), then G is locally two-transitive on S. Conversely, if G is locally two-transitive on an arbitrary thick rank-two geometry S and Γ(S) is a geometry, then S is a partial linear space. For a partial linear space, chamber transitivity is exactly transitivity on (B2).

Proof. Given an ordered pair of distinct lines M,L through x, choose P∈M∖{x} to obtain (B2). Chamber transitivity then proves that Gx is transitive on ordered pairs of its incident lines. Given an ordered pair of distinct points P,x on M, choose a different line L through x; the same argument proves that GM is two-transitive on the points of M.

For the converse, a flag object (x,M) extends to a chamber of Γ(S), so some line L through x satisfies M∩L={x}. The local two-transitivity at x makes the stabilizer of x,M transitive on all other lines through x. Hence every other such line has singleton intersection with M. Any two distinct lines therefore meet in at most one point, which is partial linearity. The last assertion is precisely the equivariant bijection (B2). ■

Projective planes, including order two

Theorem B.ProjectivePlane. For every prime power q≥2, Γ(PG(2,q)) is a firm residually connected geometry satisfying $(IP)_2$. Its original-point and original-line residues are respectively a dual affine plane and an affine plane of order q; its flag residues are Kq,q. The group PSL(3,q) is chamber-transitive.

Proof. This is the Steiner construction S(2,q+1,q²+q+1). Its parameter r=q+1 satisfies the firmness criterion. Equivalently, the three panel sizes can be read directly from (B1): q+1 completions of an anti-flag P,L, q choices of a new line through x different from M, and q points of M other than x. All are at least two.

At an original line L, the remaining points are the projective points outside L. Every other projective line M meets L in a unique point x and hence supplies exactly one flag object (x,M). The resulting incidence is the affine plane obtained by deleting L. At an original point p, the flag objects are indexed by all other projective points x, and the remaining original lines avoid p. Dualizing the projective plane gives the corresponding dual affine plane. At a flag (x,M), the remaining points are M∖{x}, the remaining lines are the q other lines through x, and every cross pair is incident in Γ(S). These are precisely the asserted rank-two residues, and they are connected.

The full incidence graph is connected. Two original points have an original line containing neither, and two original lines have an original point on neither: the complement in either count has q²−q>0 elements. Thus the subgraph on the first two types is connected, and every flag object has a neighbor there. This proves residual connectivity. The affine residues are partial linear spaces and the flag residues generalized digons, proving $(IP)_2$.

For transitivity choose projective coordinates sending x to $\langle e_1\rangle$, P to $\langle e_2\rangle$, M to $\langle e_1,e_2\rangle$ and L to $\langle e_1,e_3\rangle$. Every configuration (B2) admits such a projective frame. A linear map between two frames can be made determinant one by rescaling the third basis vector; this does not change any of the four displayed projective objects. Therefore SL(3,q), and its projective image PSL(3,q), acts transitively. ■

This proves L092 Theorem3 and includes q=2, which is required for the derived base of S(3,4,8). The source's restricted q≠2 statement alone would not justify that application. The object counts are

$$q^2+q+1, \quad q^2+q+1, \quad (q^2+q+1)(q+1),$$

and the panel sizes q+1,q,q recover its local orders and affine-residue diagram.

Affine planes

Theorem B.AffinePlane. For every prime power q≥3, Γ(AG(2,q)) is a firm residually connected geometry satisfying $(IP)_2$. The affine special linear group $\mathbb{F}_q^2 \rtimes SL(2,q)$ acts regularly on its chambers. In particular every group with the translation subgroup and linear part containing SL(2,q) is chamber-transitive.

Proof. The linear space is $S(2,q,q^2)$, with r=q+1. The three panel sizes are q,q,q−1, so the construction is firm. Its flag residues are Kq−1,q.

The point residue at the origin is the affine plane with the origin and every line through it removed. Its point-type objects are all nonzero vectors, and its line-type objects are the lines avoiding the origin. Replace each remaining line by the clique on its points. Two nonzero vectors are adjacent in this graph exactly when they lie on different rays through the origin. It is the complete (q+1)-partite graph with parts of size q−1, so it is connected. Every remaining line has q points, and every point belongs to q remaining lines; the incidence graph is therefore connected as well.

For the line residue, take L:y=0. Its point objects are (x,y) with y≠0, and its line objects are x=ay+b, with their point on L removed. Two points at different nonzero y-levels lie on a unique remaining line. If they have the same y-level, choose any point at a different nonzero level and use two such joining lines. This is possible because q−1≥2. Hence the point objects lie in one component, and every remaining line has q−1≥2 points, so the whole residue is connected.

The subgraph on original points and original lines, with nonincidence as adjacency, is connected: a line can avoid any two prescribed points, and a point can avoid any two prescribed lines, by q≥3. Every flag object attaches to this subgraph. Together with the already connected rank-two residues this proves residual connectivity. The two kinds of point/line residues are subincidence structures of the affine plane and hence are partial linear spaces; the flag residues are generalized digons.

For transitivity translate x in (B2) to zero. The nonzero vector u=P−x determines M. Choose any vector v on the other direction L. Given a second chamber, a linear map can send u to its prescribed target and the line ⟨v⟩ to its prescribed target; rescale v's image to make the determinant one. This gives chamber transitivity under $\mathbb{F}_q^2 \rtimes SL(2,q)$.

A chamber stabilizer in that group fixes zero, fixes u, and preserves a complementary one-dimensional subspace. In the basis (u,v), its matrix is diagonal with first entry one; determinant one forces the second entry to be one. Thus it is trivial. The action is regular, and there are $q^3(q^2-1)$ chambers. ■

The source's extra group at q=4 also has a complete elementary proof.

Proposition B.SemilinearSixteen. AΓL(1,16), acting on $\mathbb{F}_{16}$ as a two-dimensional affine space over F4, is regular on the chambers of Γ(AG(2,4)).

Proof. Translate the center x of (B2) to zero and multiply by the inverse of P−x to make P=1. Then M is the direction $\mathbb{F}_4$. The other line L has one of the four remaining directions, indexed by the nonidentity elements of the cyclic quotient $\mathbb{F}_{16}/F4$, of order five. The stabilizer of 0 and 1 in AΓL(1,16) is generated by the Frobenius $a \mapsto a^2$. On the direction quotient it acts by multiplication by two on exponents modulo five, a four-cycle on the nonzero exponents. Hence it is regular on the four possible L. The whole action is transitive, and its order 16·15·4=960 equals the chamber count $4^3(4^2-1)=960$. ■

These statements pay every group-action clause of L092 Theorem4.

The two affine residue families

Put Tq=Γ(AG(2,q))p and Sq=Γ(AG(2,q))L, with the identifications just proved.

In Tq, the nonzero vectors on a ray form an intrinsic noncollinearity class. For an anti-flag (P,N), the number of points of N collinear with P is q if N is parallel to the ray 0P, and q−1 otherwise. In the latter

case N meets that ray in exactly one retained point. Both cases occur for $q \geq 3$, so Tq is not a partial geometry.

In Sq, use coordinates with L:y=0. A line N has $q-1$ retained points. For an anti-flag (P,N), exactly one of those points has P's y-coordinate, and is not collinear with P in the retained geometry. All the others are joined to P by nonhorizontal lines. Thus the partial-geometry parameter is constantly $q-2$.

For $q>3$, both families have point- and line-diameter four and gonality three, as claimed in the source. The anti-flag counts just computed show that every point-line distance is at most three. Each family has noncollinear point pairs and disjoint line pairs, so both same-type diameters are exactly four. In Tq the points (1,0),(0,1),(1,1) give a proper triangle. In Sq take (0,1),(0,t),(1,s), where 1,t,s are three distinct nonzero field elements; all three joining lines remain, and the triangle is proper. Partial linearity then gives gonality three.

For q=3, Sq instead is the incidence geometry of the complete bipartite graph K3,3: its two nonzero y-levels give the two sets of three vertices, and each remaining affine line joins one vertex from each level. Its incidence graph is the subdivision of K3,3, with point/line diameters four and gonality four. This accounts for the exceptional rank-two residue in the q=3 diagram of Theorem4.

Theorem B.PointIteration (L092 Theorem5). For every $q \geq 3$, $\Gamma(Tq)$ is not chamber-transitive under its type-preserving automorphism group.

Proof. For a point P of Tq, let R(P) be P together with the other points not collinear with it. This is exactly its nonzero ray through the deleted origin and is defined by the incidence structure alone.

Every type-preserving automorphism of $\Gamma(Tq)$ restricts on its original point and line types to an automorphism of Tq, because original incidence is the complement of their incidence in (B1). Therefore it preserves the integer $|N \cap R(P)|$ for a flag (P,N) of $\Gamma(Tq)$.

There are two possible values, zero and one, and both occur. Set P=(1,0). The line N:b=1 avoids R(P), giving zero. For any $c \in Fq^* \setminus \{1\}$, the line N:a=c does not contain P and meets R(P) at (c,0), giving one. Such c exists because $q \geq 3$.

Both flags extend to chambers. For any anti-flag P,N, there are q net lines through P; excluding a possible one parallel to N still leaves at least $q-1$ choices meeting N. Their intersection x and the chosen line M give the required flag object (x,M). Thus chamber transitivity would imply transitivity on these two-object flags, contrary to the preserved integer. ■

This proof does not rely on the source's asserted full automorphism-group identification for Tq. It also avoids an ambiguity about "paths of length four."

The second iteration $\Gamma(Sq)$ is treated in the next section. Its exceptional A-geometry is q=8. The printed value q=4 fails residual connectivity; a finite Lean certificate verifies a nonconstant invariant of its literal residue graph, while the written proof below identifies the three hexagons.

Higher-dimensional spaces

Theorem B.SpaceObstruction (L092 Theorems7–8, with characteristic restrictions removed). Let $n \geq 3$. For every prime power q, $\Gamma(PG(n,q))$ is not residually connected. For every prime power $q \geq 3$, $\Gamma(AG(n,q))$ is not residually connected.

Proof. Fix an original line L. Every point P outside L determines a unique plane $\Pi = \langle L,P \rangle$. Every flag object (x,M) in the residue at L has $M \cap L = \{x\}$, so $M \neq L$ and also determines a unique plane $\Pi = \langle L,M \rangle$. If P and (x,M) are incident in the residue, then $P \in M \setminus L$ and the two planes coincide. Hence every edge, and every path, preserves Π.

When $n \geq 3$ there are at least two planes through L, each with points outside L. The residue is therefore disconnected. This already proves both assertions.

More precisely, in the projective case each component is the affine plane obtained by removing L from a projective plane Π. In the affine case each component is the deleted-line affine residue Sq inside Π, proved connected above for q≥3. The number of components in either case is

$$\frac{q^{n-1}-1}{q-1},$$

the number of one-dimensional subspaces in the quotient direction space. ■

The projective assertion includes q=2 and is the negative base needed by the binary Steiner family. The affine assertion gives the negative base for every subline design whose derived affine dimension exceeds two. Neither application needs to redo a higher-rank connectivity calculation.

Iterating the deleted-line residue

Write $q = p^e > 3$, and let L be the line $y = 0$ in $AG(2, q)$. The source's rank-two residue

$$S_q = \Gamma(AG(2,q))_L$$

has the following literal description. Its points are (x, y) with $y \neq 0$. Its lines are

$$M_{a,b} : \; x = ay + b \qquad (a, b \in \mathbb{F}_q),$$

with their points on L removed. Each line has $q - 1$ points, and each point belongs to q lines. This is exactly the residue of Construction 3: a flag (z, M) incident to L has $M \cap L = \{z\}$, so its line M is nonparallel to L and its intersection z is uniquely determined.

Pass to the dual $N_q = S_q^*$. Its points are the pairs $(a, b) \in \mathbb{F}_q^2$. For an original point (x, y) with $y \neq 0$, its dual line is

$$b = -ya + x.$$

Thus N_q is the affine plane with both the horizontal and the vertical parallel classes omitted. Its lines are exactly

$$\ell_{m,c} : \; b = ma + c \qquad (m \in \mathbb{F}_q^*,\; c \in \mathbb{F}_q). \tag{D1}$$

There are q^2 points and $q(q-1)$ lines. The line size is q, and the number of lines on a point is $q - 1$.

We use Construction3 literally. For any partial linear space T, its objects are the points P, lines ℓ, and incident pairs $F = (x, M)$ of T, with

$$P \sim \ell \iff P \notin \ell, \quad P \sim (x, M) \iff P \in M \setminus \{x\}, \quad \ell \sim (x, M) \iff \ell \cap M = \{x\}. \tag{D2}$$

The construction commutes with duality, exchanging its first two types. To verify the duality assertion of the source's Lemma 2, send an old point to its dual line, an old line to its dual point, and (x, M) to (M^*, x^*). The first relation in (D2) is symmetric under this exchange. The second becomes the assertion that two dual lines meet in the indicated dual point, which is equivalent to uniqueness of the original line through two distinct points. The third becomes membership of the indicated original point in a distinct original line. These equivalences use precisely the partial-linearity assumption. Hence

$$\Gamma(S_q) \cong \Gamma(N_q). \tag{D3}$$

Its full automorphism group

All automorphism groups in this note preserve the displayed types.

Proposition D.Aut. Every automorphism of N_q is of one of the forms

$$(a, b) \longmapsto (\alpha a^\sigma + u,\; \beta b^\sigma + v) \quad \text{or} \quad (\alpha b^\sigma + u,\; \beta a^\sigma + v), \tag{D4}$$

where $\alpha, \beta \in \mathbb{F}_q^*$, $u, v \in \mathbb{F}_q$, and $\sigma \in \mathrm{Gal}(\mathbb{F}_q/\mathbb{F}_p)$. In particular

$$|\operatorname{Aut}(S_q)| = |\operatorname{Aut}(N_q)| = 2e\, q^2(q-1)^2. \tag{D5}$$

Restriction also gives

$$\operatorname{Aut}(\Gamma(N_q)) \cong \operatorname{Aut}(N_q). \tag{D6}$$

Proof. Two distinct points of N_q fail to be collinear exactly when they have the same first coordinate or the same second coordinate. The noncollinearity graph is therefore the Hamming graph on $\mathbb{F}_q^2$. Its maximal cliques are exactly the rows and columns: once two vertices of a clique have the same first coordinate and different second coordinates, a vertex adjacent to both must share that first coordinate. The assertion with the coordinates reversed is identical.

The rows form one family of q pairwise disjoint maximal cliques, and the columns the other; every row meets every column in one point. Consequently an automorphism either permutes each family or exchanges the two. After an optional coordinate exchange, its point permutation has the form

$$(a, b) \longmapsto (\phi(a), \psi(b))$$

for arbitrary permutations ϕ, ψ of the field.

Translations and independent nonzero coordinate scalings are actual automorphisms of (D1). Compose with such maps to arrange

$$\phi(0) = \psi(0) = 0, \qquad \phi(1) = \psi(1) = 1.$$

The diagonal is then fixed, being the unique line through $(0, 0)$ and $(1, 1)$; hence $\phi = \psi = \theta$.

Parallelism of lines is intrinsic as disjointness. Since the diagonal is fixed, its entire parallel class is preserved. The line $b = a + c$ must therefore map to $b = a + \theta(c)$, its intercept being the image of $(0, c)$. It follows that

$$\theta(a + c) = \theta(a) + \theta(c).$$

Likewise the line $b = ma$ maps to the line through $(0, 0)$ and $(1, \theta(m))$, giving

$$\theta(ma) = \theta(m)\theta(a).$$

These formulas also hold at zero. Thus θ is a field automorphism. This proves necessity in (D4); substitution in (D1) proves sufficiency. The parameters are unique, and the exchanged and unexchanged forms are distinct for $q > 2$. Counting them gives (D5).

Finally any type-preserving automorphism of $\Gamma(N_q)$ restricts on its first two types to an automorphism of N_q, since original incidence is the complement of their incidence in (D2). The point-neighbors of a third-type object (x, M) form $M \setminus \{x\}$. This set contains at least two points, which uniquely recover M, and its complement in M then recovers x. Thus restriction is injective. Every automorphism of N_q plainly extends to its flags while preserving (D2), proving surjectivity and (D6). ■

The source's preceding assertion on p.568 identifies $\operatorname{Aut}(S_q)$ with the affine-plane stabilizer of L, whose order is $eq^2(q-1)^2$. Proposition D.Aut exhibits the missing factor two. In the dual model it is the exchange of coordinates. This exchange already exists before Construction3 is applied a second time.

Firmness and rank-two incidence

Proposition D.Firm. For every $q > 3$, $\Gamma(N_q)$ is a firm geometry and satisfies $(IP)_2$. It has

$$q^2, \quad q(q-1), \quad q^2(q-1)$$

objects of its three respective types, and

$$q^2(q-1)^2(q-2) \tag{D7}$$

chambers.

Proof. Every incident pair in (D2) extends to a chamber. For $P \notin \ell$, choose a line M through P with slope different from that of ℓ. There are exactly $q-2$ such choices; each intersects ℓ at a unique point $x \neq P$ and supplies the flag (x, M). For $P \in M \setminus \{x\}$, choose any of the $q-2$ lines through x other than M. For $\ell \cap M = \{x\}$, choose any of the $q-1$ points of $M \setminus \{x\}$. Thus the three types of incident pair lie in respectively $q-2$, $q-2$ and $q-1$ chambers. All these numbers are at least two. Single objects and the empty flag extend because each object has incident pairs. This proves both the geometry axiom and firmness.

A chamber is uniquely specified by (x, M), a point $P \in M \setminus \{x\}$, and a line $\ell \neq M$ through x. The three choices have sizes $q^2(q-1)$, $q-1$ and $q-2$, proving (D7).

The residue of a flag object (x, M) is the complete bipartite graph on the $q-1$ points of $M \setminus \{x\}$ and the $q-2$ other lines through x. For a point residue, its flag objects correspond to the distinct points collinear with that point, and a pair of them lies on at most one line of N_q. Thus this residue is a partial linear space. The analogous statement for a line residue follows by duality, or directly from the fact that two distinct lines meet in at most one point. Hence every rank-two residue is either a partial linear space or a generalized digon, exactly $(IP)_2$. ■

Residual connectedness

For a partial linear space T and point z, form the graph on the points collinear with z, with an edge between x, y when x, y, z are pairwise collinear but are not on one line. In the point residue $\Gamma(T)_z$, its flag objects are naturally these collinear points: (x, M) is uniquely determined by x because M is the unique line through x, z. Its other objects are the lines not on z. Such a line is incident to all its points collinear with z. Consequently the displayed graph is the graph obtained by replacing each of these line objects by a clique on its neighbors. Provided no such line object is isolated, connectivity of the two graphs is equivalent. We apply this elementary reduction to N_q and its dual, paying the non-isolation condition explicitly below.

Proposition D.Connected. For $q > 3$, $\Gamma(N_q)$ is residually connected if and only if $q > 4$. At $q = 4$, each residue of an original point of N_q is the disjoint union of three hexagons.

Proof. First take the original point $z = (0, 0)$ of N_q. Its collinear points are

$$(a, b) \in (\mathbb{F}_q^*)^2.$$

Two such points are adjacent in the preceding graph exactly when

$$a \neq a', \qquad b \neq b', \qquad b/a \neq b'/a'. \tag{D8}$$

The first two inequalities give their collinearity in the net, and the last says that their joining line does not contain the origin. Every net line avoiding the origin has exactly $q-2$ points collinear with the origin: its intersections with the horizontal and vertical axes are distinct, and all its other points have both coordinates nonzero. Thus no line object of the residue is isolated.

Choose a generator ζ of $\mathbb{F}_q^*$ and put $n = q-1$. In exponents, (D8) is the Cayley graph of

$$(\mathbb{Z}/n\mathbb{Z})^2$$

whose allowed differences (r, s) satisfy $r \neq 0$, $s \neq 0$ and $r \neq s$. If $n \geq 4$, both $(1, 2)$ and $(1, 3)$ are allowed. Their difference is $(0, 1)$ and subtracting twice this vector from $(1, 2)$ gives $(1, 0)$. They therefore generate the whole group, so the graph and the original point residue are connected.

For $q = 4$, one has $n = 3$. The only allowed differences are $(1, 2)$ and $(2, 1)$, which generate the subgroup of order three on which the sum of coordinates is zero. The graph has three components, each a triangle. Every line object of the incidence residue has degree two, and each pair of adjacent point objects has a unique joining line. Each triangle therefore lifts to a hexagon. There are exactly three disjoint hexagons, not one connected residue.

Next take an original line of N_q. By the coordinate automorphisms it suffices to use the diagonal $\ell : b = a$. In the dual near-point graph its vertices are the lines of N_q meeting ℓ, other than ℓ itself. Write such a line by its slope $m \in \mathbb{F}_q^* \setminus \{1\}$ and its intersection (t, t) with ℓ. Two vertices $(t, m), (t', m')$ form a proper dual triangle with ℓ exactly when

$$t \neq t' \quad \text{and} \quad m \neq m'. \tag{D9}$$

If the slopes differ their intersection exists; it lies on ℓ precisely when the two displayed intersection points agree. Thus this graph is the direct product $K_q \times K_{q-2}$. It is connected for $q > 3$. One elementary verification is that two vertices agreeing only in the second coordinate have a two-edge path through a new first coordinate and a different second coordinate. For vertices agreeing only in the first coordinate the same argument applies if $q - 2 \geq 3$; when $q - 2 = 2$, a three-edge path using two other first coordinates suffices.

The original point objects in this line residue correspond to points outside ℓ. Through each such point there are $q - 2$ net lines meeting ℓ: exclude only the parallel direction. Hence the clique expansion has no isolated omitted objects, and the line residue is connected.

The remaining rank-two residues are the complete bipartite flag residues from Proposition D.Firm, so they are connected. The full incidence graph is connected as well. Indeed any two original points have an original line avoiding both, since at most $2(q-1)$ of the $q(q-1)$ lines contain either point. Any two original lines have an original point outside their union, since $q^2 > 2q$. Thus the subgraph on the first two types is connected, and every flag object has neighbors there. These are all residues of rank at least two in a rank-three geometry, completing the proof. ■

Under (D3), the disconnected residues at $q = 4$ are exactly the original-line residues of $\Gamma(S_4)$. The source's claimed exceptional geometry has 12,16,48 objects and 288 chambers, but its sixteen original-line residues each have three hexagonal components. Its failure is residual connectedness, independently of any group-order calculation.

Flag transitivity

Proposition D.Transitivity. For $q = p^e > 3$, $\Gamma(N_q)$ is flag-transitive if and only if $q = 4$ or $q = 8$.

Proof. A chamber can be encoded as the ordered configuration

$$P \in M, \quad x \in M, \quad P \neq x, \quad x \in \ell, \quad M \neq \ell.$$

The group (D4) can translate x to $(0, 0)$ and independently scale the two coordinates to send P to $(1, 1)$. Both coordinate differences are nonzero because M is a net line. This also sends M to the diagonal. The remaining line ℓ then has an arbitrary slope

$$\lambda \in \mathbb{F}_q^* \setminus \{1\}.$$

The stabilizer of this normalized ordered data consists exactly of simultaneous field automorphisms, optionally followed by coordinate exchange. It acts on λ by

$$\lambda \longmapsto \lambda^{p^j}, \qquad \lambda \longmapsto \lambda^{-p^j}. \tag{D10}$$

This follows directly from (D4): fixing the origin removes both translations, and fixing $(1, 1)$ sets both scaling constants equal to one. Hence chamber transitivity is equivalent to transitivity of the action (D10).

Its acting group has order dividing $2e$. Therefore a necessary condition is

$$q - 2 \mid 2e. \tag{D11}$$

For prime powers $q > 3$, this leaves only $q = 4, 8$. For $p \geq 5$, already $p^e - 2 > 2e$ for all $e \geq 1$. For $p = 3$, the range $q > 3$ gives $e \geq 2$, and $3^e - 2 > 2e$. For $p = 2$, all $e \geq 4$ satisfy $2^e - 2 > 2e$. These inequalities follow from their first case by induction. The two remaining powers of two are four and eight.

For $q = 4$, inversion exchanges the two elements of $\mathbb{F}_4^* \setminus \{1\}$. For $q = 8$, choose a generator ζ of the multiplicative group of order seven. The six exponents

$$1,\ 2,\ 4,\ -1,\ -2,\ -4 \pmod 7$$

are all the nonzero residues modulo seven. Thus the Frobenius and inversion orbit of ζ is exactly $\mathbb{F}_8^* \setminus \{1\}$. This proves sufficiency in both cases. By Proposition D.Aut these are the full type-preserving automorphisms of the constructed geometry, so no additional chamber transitivity was omitted. ■

The corrected exceptional value

Theorem D.DeletedLine. Let $q > 3$ be a prime power, let L be a line of $AG(2, q)$, and put $S_q = \Gamma(AG(2, q))_L$ using the source's Construction3. Then:

1. $\Gamma(S_q)$ is a firm geometry satisfying $(IP)_2$ for every such q.
2. It is residually connected exactly when $q > 4$.
3. It is flag-transitive exactly when $q = 4$ or $q = 8$.
4. Consequently it is an A-geometry exactly when $q = 8$.

At $q = 8$ the three type sizes, in the source's original point/line/flag order, are 56,64,448. There are 18,816 chambers, and its full type-preserving automorphism group has order 18,816 and acts regularly on them. The group is the dual of the semiaffine group in (D4).

Proof. Combine (D3) and the four preceding propositions. The count and regularity assertions follow from (D5), (D7) and transitivity. ■

Hermitian unitals

Let $K = \mathbb{F}_{q^2}$ and let V be a nondegenerate three-dimensional Hermitian space. Its isotropic one-spaces, with the intersections of projective secants as blocks, form the classical 2-$(q^3+1, q+1, 1)$ unital $U_H(q)$. Each point belongs to q^2 blocks. These standard design parameters are the same ones used in [L].

Theorem H.Unital. For every prime power $q \geq 2$, $\Gamma(U_H(q))$ is firm and residually connected and satisfies $(\mathrm{IP})_2$. The group $PGU(3, q)$ acts regularly on its chambers.

Proof. The parameter $r = q^2$ satisfies the hypotheses of [G, Theorems S.1 and S.6], proving firmness and the rank-two intersection property. Directly, the three panel sizes for point–block, point–flag and block–flag pairs are $q + 1$, $q^2 - 1$ and q. At a flag (p, M) the residue is K_{q,q^2-1}. At a point Q, its flag objects are (p, Qp) for $p \neq Q$ and its block objects are the blocks avoiding Q; incidence is $p \in L$. If p, r lie on different blocks through Q, their joining block avoids Q and connects them. If they lie on the same such block, choose s outside it and use ps, sr. Every remaining block has point neighbors, so the point residue is connected. The block residue is connected by [G, Theorem H.3], which identifies it with the full isotropic-direction graph on a nonzero norm shell. All field, transport and generation arguments for this step are proved in that common paper.

The whole incidence graph is connected. Given distinct original points Q, R, choose p outside their block. At most two of the q^2 blocks through p contain Q or R, leaving one incident in Γ with both. Every block

has an outside point and every flag (p, M) has a neighbor in $M \setminus \{p\}$. These are all residues of rank at least two, proving residual connectedness.

For the action, a chamber is $(Q, L, (p, M))$ with $Q \neq p$ on M and $p \in L \neq M$. The unitary group is transitive on ordered distinct isotropic points: choose representatives e, f, rescale their nonzero pairing to one, complete by a vector of norm one orthogonal to both, and map one such basis to another. Fix $p = \langle e_1 \rangle$ and $Q = \langle e_3 \rangle$ in coordinates with

$$h(x, x) = x_1 x_3^q + x_1^q x_3 + x_2^{q+1}.$$

All projective lines through p other than $M = pQ$ have the form $L_t = \langle e_1, e_2 + te_3 \rangle$, $t \in K$. The line L_0 is tangent. For $t \neq 0$ the equation $xt^q + x^q t + 1 = 0$ has q solutions, so L_t is a secant. The projective stabilizer of (p, Q) is exactly

$$D_a = \mathrm{diag}(a, 1, a^{-q}), \qquad a \in K^*.$$

Indeed any such isometry preserves the two isotropic lines and their one-dimensional orthogonal complement. Its scalar on the latter has norm one; dividing by that scalar gives the displayed matrix. It sends L_t to $L_{ta^{-q}}$, regularly on the $q^2 - 1$ possibilities. Thus $PGU(3, q)$ is chamber-transitive with trivial chamber stabilizer. The chamber count is $q^3(q^3 + 1)(q^2 - 1)$, also obtained from [G, Theorem S.3]. □

For $q > 2$ we now recover the diameter and gonality data in Figure 2 of [L]. For a bipartite incidence graph, (d_0, g, d_1) denotes the two vertex-type eccentricities and half its girth, using the source's convention. Throughout q>2. For a bipartite incidence graph, write (d_0,g,d_1) for the two vertex-type diameters and half its girth, in the source's convention.

Theorem H.Labels. In the literal Construction3 on UH(q), the original-point residue has parameters (4,3,4). The original-block residue, ordered by outside points and flag objects, has parameters

$$(5, 4, 6) \quad \text{if q is even}, \qquad (6, 4, 6) \quad \text{if q is odd}. \tag{H1}$$

The original-flag residue is K(q,q^2-1).

The Cayley graph

Let W be the Hermitian plane in Part III of the common paper and let

$$X = \{v \in W : h(v, v) = -1\}.$$

Its points form the point type of the block residue. Two distinct points v,w are collinear exactly when h(v−w,v−w)=0. The group SU(W) acts regularly on X, as proved in Theorem H.1 of the common paper.

Fix $v_0 \in X$ and choose an orthogonal basis beginning with v_0 and with both vectors of norm−1. A determinant-one unitary matrix has the form

$$\begin{pmatrix} a & b \\ -b^q & a^q \end{pmatrix}, \qquad a^{q+1} + b^{q+1} = 1.$$

A direct calculation gives h(gv_0-v_0,gv_0-v_0)=tr(g)−2. Consequently the point graph is the Cayley graph of SU(W)≅SL(2,q) whose connection set consists of all nonidentity trace-two matrices. These are precisely the nonidentity unipotents. The conjugation in the proof of Theorem H.2 preserves traces.

Every noncentral element of SL(2,q) has distance at most two from1. This is immediate for trace two. Otherwise conjugate the element to g=(a b;c d) with c≠0; such a basis exists because a nonscalar transformation does not preserve every one-dimensional subspace. Put

$$u = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad s = \frac{\mathrm{tr}(g) - 2}{c} \neq 0.$$

Then tr($u^{-1}g$)=2 and $u^{-1}g \neq 1$, so g is a product of two connection elements. In even characteristic the only central element is1, and the diameter is two. In odd characteristic -1 is not a product of two connection elements: if $-1=uv$ and tr(u)=2, then tr(v)=tr($-u^{-1}$)=$-2\neq 2$. It is a product of three, since $-u^{-1}$ is noncentral and hence a product of two. Thus the diameter is three, and the only point at distance three from a given point is its antipode.

The incidence graph has no four-cycles by partial linearity. It also has no six-cycles. To see this, normalize three pairwise adjacent points to1,u,v. Conjugate u to (1 s;0 1), $s\neq 0$. From tr(v)=tr(u^{-1}v)=2, the lower-left entry of v is zero, and its determinant then forces both diagonal entries to be 1. Hence u and v lie in the same upper root group. In the norm-shell coordinates this root group moves points along a single isotropic affine direction, so all three points lie on the same flag object. They cannot form a six-cycle using three distinct flag objects.

There is an eight-cycle. Take nonidentity upper and lower unipotents u,v. The four points1,u,uv,v form a quadrangle in the point graph: their successive quotients are unipotent, whereas tr(uv) and tr(u^{-1}v) differ from2. Its diagonals are therefore absent. Its four joining flag objects are distinct, producing an eight-cycle in the incidence graph. The gonality is exactly four.

The two incidence diameters

Every flag object has q neighboring points. In odd characteristic at most one of them can be the antipode of a prescribed point; in even characteristic every point is within point-graph distance two. Thus every point–flag distance is at most five, in either characteristic.

Some point–flag distance is five. A fixed point has q+1 neighboring flags. Since the incidence graph has girth eight, its flags at distance three number exactly (q+1)(q−1)q. Hence only

$$(q+1)(1+q(q-1)) = q^3+1$$

flags lie within distance three, whereas the total number is (q+1)(q^2-1). Their difference is (q−2)(q+1)>0. Combined with the point-graph diameters, this gives point-type diameter five for even q and six for odd q.

Any two flag objects are at distance at most six: choose a point on the first and a point on the second which is not its antipode, and use the point-graph bound. To see that six occurs, use hyperbolic coordinates

$$h((x,y),(x,y)) = xy^q + x^q y.$$

Fix $b\neq 0$ and $c\in Fq^*\setminus\{1\}$. The shell lines F_b and F_{cb}, with constant second coordinates b and cb, are distinct flag objects of the same isotropic direction. Each contains q points. For (x,b)∈F_b and (x′,cb)∈F_{cb}, the shell equations give

$$h((x,b)-(x',cb),(x,b)-(x',cb)) = c+c^{-1}-2 = (c-1)^2/c \neq 0.$$

Thus no point of one flag is collinear with a point of the other. The flags have neither a common point nor a path of length four; their distance is six. This proves (H1).

The original-point residue

Fix the deleted unital point Q. The remaining points split into q^2 classes of q points, one class for each block through Q. Two remaining points are collinear in this residue exactly when they belong to different classes. Its point graph is therefore complete multipartite, and the largest point–point distance in the incidence graph is four.

A remaining block meets q+1 different classes. Every point not on that block is collinear with at least one of its points, so every point–block distance is at most three. Two blocks have distance at most four by

the same argument. Disjoint remaining blocks exist: fix one block L and choose $x \notin L \cup \{Q\}$; at most $q+1$ blocks through x meet L, and at most one further block contains Q, whereas $q^2 > q+2$. Thus the line-type diameter is four as well.

Choose x,y in different Q-classes, and choose z outside the union of Qx,Qy and xy. Such z exists since $q^3+1 > 3(q+1)$. These three points give a proper triangle in the residual linear space. Partial linearity excludes four-cycles, so its gonality is three. This proves (4,3,4). The complete-bipartite flag residue was proved above.

Comparison with the printed diameter labels

The right-hand label of every literal row is (4,3,4). The left-hand label is (6,4,6) for q=3,5,7,9,11,13 and (5,4,6) for q=4,8,16. Therefore the left q=5 and q=8 labels and the right q=5,q=8,q=11 labels in the printed Figure2 disagree with the literal construction. All remaining diameter/gonality labels agree. The following count records the panel discrepancies at q=5,8,11 and the object-count discrepancies at q=16.

Together these formulas determine every numerical label in Figure 2. For the literal Construction3 on the classical unital UH(q), the design parameters are 2-(q^3+1,$q+1$,1), and each point is on q^2 blocks. Therefore:

- point objects: q^3+1;
- marked flag objects: $q^2(q^3+1)$;
- block objects: $q^2(q^2-q+1)$;
- panel sizes, in Figure2's point/marked/block order: q, $q+1$, q^2-1;
- local orders printed under those nodes should be $q-1$, q, q^2-2.

These are direct counts of the actual source incidences. They do not depend on the claimed all-q connectivity or on a choice of acting group.

Panel sizes

Three printed rows have smaller marked/block panel sizes:

q	Printed local orders	Literal local orders
5	4,1,7	4,5,23
8	7,2,20	7,8,62
11	10,3,39	10,11,119

In each of these rows, the printed marked panel size is $(q+1)/3$ and the printed block panel size is $(q^2-1)/3$. Thus the diagram cannot describe the full literal Construction3 with the printed object set.

This factor three agrees with [PGU(3,q):PSU(3,q)]=gcd(3,q+1) in precisely these rows. A selected chamber orbit under the smaller group can retain the transitive object sets while omitting incidences of the complete construction. That is a possible explanation of the table, not a verified reconstruction of the source's computational procedure. No claim that the author used a particular algorithm is made here.

Object counts at q=16

The printed row gives 1,028 point objects, 263,206 marked objects and 15,482 block objects. Literal unital parameters instead give:

Object type	Printed	Literal Construction3
Points	1,028	4,097
Marked flags	263,206	1,048,832
Blocks	15,482	61,696

The printed row is internally incompatible with its own q=16 flag valencies: 1,028·256=263,168 and 15,482·17=263,194, neither equal to 263,206. This is a numerical correction independent of any interpretation of the group-index issue above.

All other printed object counts for q=3,4,5,7,8,9,11,13 agree with the literal formulas. The local orders in rows q=3,4,7,9,13,16 agree with $q-1, q, q^2-2$.

Projective sublines

Let $q \geq 3$, $e \geq 2$, $Q = q^e$, and let $\mathcal{S}$ consist of $\Omega = PG(1, Q)$ and the $PGL(2, Q)$ images of $PG(1, q)$. This is $S(3, q+1, Q+1)$. To verify uniqueness, carry an ordered triple to $(\infty, 0, 1)$. Any image subline containing that triple can be parametrized by a projectivity whose three preimages lie in $PG(1, q)$; compose with its unique $PGL(2, q)$ normalization. The resulting projectivity fixes three points and is the identity. Thus the unique block containing the normalized triple is the standard subline. The number of blocks through a pair is

$$r = \frac{Q-1}{q-1}.$$

In particular the parameters satisfy the firm-construction hypotheses of [G, Theorem S.1].

Theorem I.Subline. The geometry $\Gamma(\mathcal{S})$ is firm and satisfies $(\mathrm{IP})_2$ for every displayed q, e. It is residually connected if and only if $e = 2$. Among these connected geometries it is flag-transitive under its full type-preserving automorphism group if and only if $q = 4$.

Proof. Firmness and $(\mathrm{IP})_2$ follow from [G, Theorems S.1 and S.6]. The derivative at ∞ is $AG(e, q)$: deleting ∞ from a block through it gives $a\mathbb{F}_q + b$, $a \neq 0$. Conversely every subline through ∞ admits a representing projectivity fixing ∞, by composing with a member of $PGL(2, q)$; it is therefore of this form. Projective transitivity gives the same derivative at every point. The common residue theorem S.5, Theorem B.AffinePlane and Theorem B.SpaceObstruction now give connectedness exactly when $e = 2$. This proves the residual-connectivity assertion without identifying it with flag transitivity, as the printed Lemma 5 of [L] does.

For $e = 2$, the full design group is $P\Gamma L(2, q^2)$. Here is the precise outside input for the upper bound. The design has its sharply three-transitive $PGL(2, q^2)$ action, and the proof of [K, Theorem 3, p.73] bounds its full three-transitive group by the semilinear projective group. Its degree $q^2 + 1$ is neither a power of two nor one of the Mathieu degrees in that classification. Conversely all semilinear projectivities preserve the block family. The reconstruction theorem S.2 identifies this full design group with the full type-preserving group of $\Gamma(\mathcal{S})$.

Write $q = p^f$. The chamber formula S.3 gives

$$|\operatorname{Ch}\Gamma| = (q^2+1)q^2(q^2-1)q = |PGL(2, q^2)|q, \qquad |\operatorname{Aut}\Gamma| = 2f|PGL(2, q^2)|.$$

Transitivity requires $q \mid 2f$. If $p \geq 3$, then $p^f > 2f$ for all $f \geq 1$. If $p = 2$, the assumptions give $f \geq 2$, and $2^f \leq 2f$ only for $f = 2$. Thus $q = 4$ is necessary.

For sufficiency take $Q = 16$. Normalize a flag (x, T, B_0), $x \in T$ and $|T| = 2$, to $x = \infty$, $T = \{\infty, 0\}$, $B_0 = PG(1, 4)$. This is possible using $PGL(2, 16)$ and then the $PGL(2, 4)$ stabilizer of that block. Alternative blocks through T are $\{\infty\} \cup \alpha\mathbb{F}_4$, where $\alpha\mathbb{F}_4^*$ ranges over the four nonidentity classes of

$\mathbb{F}_{16}^*/\mathbb{F}_4^*$, a cyclic group of order five. Frobenius $z \mapsto z^2$ cycles these four classes. Once a class is chosen, the scalars in $\mathbb{F}_4^*$ act transitively on the three possible copied points in $\alpha\mathbb{F}_4^*$. Hence the stabilizer of the normalized flag is transitive on all chamber extensions, proving sufficiency. □

For completeness the six rank-two residues are described explicitly. Use types $0, 1, 2, 3$ for ordinary point, marked pair, block, copied point. Fixing types $01, 12, 13$ gives respectively $K_{r-1,q-1}$, $K_{2,q-1}$ and $K_{2,r-1}$. Fixing 23, namely $B_0, z \notin B_0$, leaves the subdivision of the complete graph on B_0: each marked pair is uniquely determined by a two-subset of B_0 and the unique block containing it together with z. Fixing 03 gives $AG(e, q)$ with one point and the lines through it removed. Fixing 02 gives its points outside a line L and lines meeting L in exactly one point. The latter identifications follow by taking the ordinary point to ∞ in the derivative just described; the marked object records the residual line and its unique intersection with L, or the residual point and its unique line through the removed point, respectively. These constructions give inverse object maps and preserve all incidences. The final residue has exactly $(q^{e-1} - 1)/(q - 1)$ connected components by B.SpaceObstruction.

The literal $S(3, 4, 10)$ example, with $q = 3, e = 2$, is a finite counterexample to the printed restriction $q = 4$ for connectedness. Its type sizes are $10, 180, 30, 10$, and it has 2160 chambers. The exact checker verifies all 1881 residues of rank at least two. This computation checks a consequence of the uniform proof; it is not the proof for arbitrary q or e.

The binary affine family

Let S_d consist of points and affine planes of AG(d,2), d≥3. Every three distinct points are affinely independent over F_2 and lie in a unique affine plane, so this is $S(3, 4, 2^d)$. Deriving at the origin identifies the remaining points with nonzero vectors. A derived block is {u,v,u+v}, precisely a projective line in PG(d−1,2). Thus the derived design is PG(d−1,2).

The projective base theorem above and Theorem S.5 of the common paper show:

$$\Gamma(S_d) \text{ is residually connected } \iff d = 3. \tag{C4}$$

This proves L092 Lemma4 without repeating an obstruction at every rank. The positive base PG(2,2) is included explicitly; the source's Theorem3 excludes q=2, so one must not cite that restricted statement alone for this application.

The full affine group is chamber-transitive for every d≥3. To apply Theorem S.3, note first that AGL(d,2) is transitive on ordered triples of distinct points: translate the first point to zero, then send the two independent difference vectors to prescribed independent vectors. Normalize a triple to $(0,e_1,e_2)$. The unique block on it is $\langle e_1,e_2\rangle$. Alternative planes through the line $\langle e_1\rangle$ correspond to nonzero vectors in

$$\mathbb{F}_2^d/\langle e_1\rangle$$

other than the vector represented by e_2. The stabilizer fixing e_1,e_2 induces the full stabilizer of that nonzero vector in GL(d−1,2). This group is transitive on the remaining nonzero vectors: any ordered pair of independent vectors extends to a basis. Theorem S.3 proves chamber transitivity.

At d=3 the type sizes are 8 ordinary points, 84 marked objects, 14 blocks and 8 copied points; the panel sizes are 2,3,2,2 in that type order. There are $(8)_3 \cdot 2 = 672$ chambers. The group AGL(3,2), of order 1344, has chamber stabilizer of order 2. These formulas give the numerical positive case in Theorem 11 of [L]. The full design group is exactly AGL(d,2). Indeed, after translating the image of zero back to zero, a design automorphism preserves the unique fourth point x+y of the affine plane through 0,x,y whenever x,y are distinct nonzero vectors. It therefore preserves addition in this case; equal vectors or a zero vector give the same identity immediately in characteristic two. Hence it is linear. This proves the full-group identification directly, while chamber transitivity already follows from the explicit subgroup exhibited above.

The Mathieu applications

For S(4,5,11) and S(5,6,12), the chamber formula of Theorem S.3 gives respectively 23,760 and 285,120 chambers. The literal finite computation `phase186_witt_derived_check.g` constructs their forced native block systems, verifies the Steiner property and computes full automorphism groups of orders 7,920 and95,040. Neither group can therefore be transitive: the chamber count is three times the full group order. No search over putative chamber orbits is necessary.

For S(3,6,22), S(4,7,23) and S(5,8,24), the common value is r=5 and the final fiber has size 4. Their chamber counts are respectively 36,960,850,080 and20,401,920. The complete program `phase182_mathieu_fibers.g` constructs the actual block designs, verifies the Steiner property over every t-subset, and checks the stabilizer orbits supplying the source's Property(*). That computation is retained; it is not duplicated here.

Their (t−2)-fold derived design is PG(2,4). This is now established by explicit coordinate-incidence maps, described below and checked by GAP, independent coefficient arithmetic, and the finite Lean certificate `WittDerivedPlanes.lean`. The bridge from the original native Witt blocks to each displayed derived incidence table remains checked in GAP and Python. The residue recursion, Theorem S.6 and the projective base theorem then prove geometry, firmness, residual connectivity and $(IP)_2$, while the retained fiber computation proves chamber transitivity.

For the rank-six M24 construction, the ordinary-point residue is Γ(S(4,7,23)), with type sizes 23,253,8,855,253,23. The residue of an octad block has type sizes 8,28,56,280,16: its ordinary subsets have sizes 1,2,3; a marked object chooses a four-subset of the octad and one of four alternative octads; and its copied points lie outside the octad. These two type-size multisets prove that the displayed rank-five residues are not isomorphic. The Figure 3 subgroup labels are proved below by explicit finite kernels, complements and transported actions. The object-level identifications with the named earlier Pasini geometries remain source-specific obligations.

Witt designs and their actual groups

The program `phase186_witt_derived_check.g` constructs the five native Mathieu designs from group orbits. For M_{11} and M_{12} the orbits on five- and six-subsets have sizes 66,396 and132,792, respectively. Thus exactly one orbit can be the block set of the required Steiner system. For each large Mathieu group, the pointwise stabilizer of a fixed ordered t-tuple has two remaining point orbits, of sizes 3 and16. The unique block through the tuple must consist of the tuple and the size-three orbit. This proves uniqueness of the native invariant block system before the actual Steiner condition is checked.

The program then checks that each t-subset occurs in exactly one block. The full design automorphism group is computed afresh from the incidence structure, its generators are checked to preserve the complete block set, and its relation to the original Mathieu subgroup is tested as an actual permutation-group inclusion or equality.

Native model	Blocks	t-subsets checked	Native group order	Full design group order
S(4,5,11)	66	330	7,920	7,920
S(5,6,12)	132	792	95,040	95,040
S(3,6,22)	77	1,540	443,520	887,040
S(4,7,23)	253	8,855	10,200,960	10,200,960
S(5,8,24)	759	42,504	244,823,040	244,823,040

For degrees11,12,23 and24 the full group equals the native group, not merely a group of the same order. At degree22 the native M_{22} is a subgroup of index two. This supplies the full-group bounds in the small-

Witt exclusions: the chamber counts23,760 and285,120 exceed the respective full group orders by a factor of three.

Explicit maps to derived projective planes

The target plane is constructed directly from GF(4)³. Points are nonzero vectors normalized so that their first nonzero coordinate is1. Lines are the kernels of the21 similarly normalized nonzero linear forms. Hence the model is PG(2,4) by its coordinate definition, without invoking uniqueness of projective planes of order 4.

For each large Witt design, remove the fixed (t−2)-set R={1,…,t−2} from every block containing R. The program finds an isomorphism of the resulting two-colored incidence graph with the coordinate plane. It then discards the graph-search conclusion as a black box and verifies the explicit point permutation: the mapped block set equals the coordinate line set, and all441 point/line incidences and nonincidences agree. Native t-transitivity transports this conclusion to every (t−2)-fold derivative.

The saved JSON witness records all original blocks, all derived blocks, the21 coordinate vectors, the coordinate lines, and each explicit point map. The independent `phase186_witt_validate.py` recomputes GF(4) arithmetic as coefficient-bit arithmetic modulo z²+z+1, reconstructs all coordinate lines, reconstructs each derivative from its original blocks, and checks each map. It also checks every Steiner condition and the design-preservation action of every saved group generator. The validator does not claim an independent proof that the saved automorphism generators exhaust the full group.

The file formal/WittDerivedPlanes.lean verifies explicit point and line bijections for all three displayed derivatives and proves that all incidences agree with the literal GF(4) dot-product equation. It reuses the frozen F4 arithmetic definitions and verified field-table laws in DeletedLineFourResidue. All nine retained axiom reports use only propext, Classical.choice and Quot.sound. There is no assumed incidence-isomorphism axiom, sorry, admit or native_decide. Two deliberately altered coordinate-incidence maps are rejected by Lean in the fresh replay. The generator binds the sorted GAP line-support convention to the separately ordered coordinate linear forms; the resulting source is checked byte-for-byte against the reviewed Lean file.

This formal certificate starts from the displayed finite derivative tables. Its identification of those tables as derivatives of the original Witt blocks remains outside Lean, checked independently by GAP and Python. The all-rank residue recursion, full design automorphism groups, and stabilizer structures also remain outside this Lean certificate.

The displayed sizes and local orders

Theorem S.3 gives the following object sizes, in the order ordinary subset types, marked objects, blocks, copied points. The panel sizes are listed in the same order; the corresponding local orders are one less. This states the numerical Figure 3 data independently of the group identifications.

Design	Object sizes	Panel sizes
$S(2,5,21)$	$105, 21, 21$	$5, 4, 4$
$S(3,6,22)$	$22, 1155, 77, 22$	$2, 5, 4, 4$
$S(4,7,23)$	$23, 253, 8855, 253, 23$	$2, 2, 5, 4, 4$
$S(5,8,24)$	$24, 276, 2024, 53130, 759, 24$	$2, 2, 2, 5, 4, 4$

Indeed the block and marked counts are respectively $\binom{v}{t}/\binom{k}{t}$ and that number times $\binom{k}{t-1}$; the ordinary counts are $\binom{v}{j}$. All four rows have $r = 5$. The coordinate-plane row is the $q = 4$ case of B.ProjectivePlane; its rank-two residues are affine planes and a generalized digon. For the three Witt rows the common residue decomposition S.4 and the explicit derived-plane maps give the same terminal

residue data, with the Boolean intervals supplying the ordinary chains. Thus the geometric meaning of the diagram is not inferred from the numerical counts alone.

Block and marked-object stabilizers

Let G be the native group, B a block of size k, and H=G_B. For k=6,7,8, restriction to B gives an exact sequence

$$1 \longrightarrow N \longrightarrow H \longrightarrow A_k \longrightarrow 1.$$

The computation proves that N is elementary abelian of order 16, the image is the actual alternating permutation group on B, and a complement C exists with N∩C=1 and NC=H. Thus H=2^4:A_k, with the action supplied by the saved permutation realization. An order factorization alone is not used to infer splitting.

A marked object is (T,B∖T), where |T|=t−1. Its stabilizer M is exactly H_T. Since N fixes every point of B, it fixes T, and

$$M = N \rtimes C_T, \qquad C_T \cong (S_{t-1} \times S_{k-t+1}) \cap A_k. \tag{W1}$$

The program verifies the actual image equality, the complement intersection and the product order in (W1). These give all block and marked labels in the three Mathieu rows of Figure3:

Group	Block stabilizer	Marked stabilizer	Orders
M_{22}	2^4:A_6	2^4:S_4	5,760;384
M_{23}	2^4:A_7	2^4:((C_3×A_4):C_2)	40,320;1,152
M_{24}	2^4:A_8	2^4:((A_4×A_4):C_2)	322,560;4,608

For the latter two descriptions, take the even permutations in the two disjoint factors. Their product has index two in the parity-matched subgroup; a pair of transpositions, one in each factor, gives a complementary involution. This proves the displayed semidirect descriptions directly.

The source writes the M_{22} marked group as $2^2{\cdot}2^4$:S_3. To bind that notation as well, the program computes Q=O_2(M), proves |Q|=64, Z(Q)≅2^2 and Q/Z(Q)≅2^4, proves M/Q≅S_3, and finds a complement to Q. Thus both descriptions refer to the same explicitly constructed group. The dot in the source's middle extension is retained; no unwarranted splitting of Q over its center is asserted.

For the first row, G=PSL(3,4) acts on the coordinate plane. Applying the identical block calculation to a line of size 5 gives the line stabilizer2^4:A_5, and fixing a point on that line gives the marked stabilizer2^4:A_4. Projective duality, which normalizes PSL(3,4), gives the same2^4:A_5 point stabilizer. Their orders are 960 and192.

Ordinary-subset stabilizers

For an ordinary j-subset R, the set stabilizer G_R acts on the j-fold derived design. The computation verifies faithfulness of this action, finds an explicit incidence map to the appropriate smaller native Witt design or coordinate projective plane, and conjugates the actual residual permutation group by that map.

The pointwise stabilizer of R becomes precisely the smaller native Mathieu group, or precisely the group generated by the projective action of SL(3,4) when j=t−2. This is a group equality in the target permutation action. The set stabilizer gives:

Native group	Ordinary subset size	Actual identified stabilizer
M_{22}	1	PSL(3,4), order 20,160
M_{23}	1	M_{22}, order 443,520

Native group	Ordinary subset size	Actual identified stabilizer
M_{23}	2	PSL(3,4):C_2, order 40,320
M_{24}	1	M_{23}, order 10,200,960
M_{24}	2	full Aut(S(3,6,22)), order 887,040
M_{24}	3	PΓL(3,4)=PSL(3,4):S_3, order 120,960

The source labels the penultimate group M_{22}:2. The current full-group computation gives the actual degree22 overgroup; identifying the colon as a split extension additionally requires a complementary involution, which is checked in the finite runner. The projective index-two case is also checked to split. For the final case the program builds PΓL(3,4) explicitly from GL(3,4) acting on normalized vectors together with the coordinate Frobenius map, and verifies equality with the transported stabilizer. Its quotient over PSL(3,4) is C_3:C_2: determinant modulo cubes gives C_3, and Frobenius inverts it. A diagonal matrix of order 3 and the Frobenius map supply a complement, proving the displayed S_3 extension.

Copied-point stabilizers are ordinary-point stabilizers. This proves the Figure 3 isomorphism labels, with explicit actions rather than recognition from orders.

The higher-rank classification

Theorem S.Classification. Let S=S(t,k,v) be a finite Steiner system with $3\le t<k$, and let G=Aut(S). Then Γ(S) is a firm residually connected geometry on which G is flag-transitive if and only if S is one of the following:

1. the points and affine planes of AG(3,2);
2. the projective GF(4)-subline design on PG(1,16);
3. one of the three large Witt designs S(3,6,22), S(4,7,23), S(5,8,24).

Each resulting geometry also satisfies $(IP)_2$. In the first case the full design group is AGL(3,2); in the second it is PΓL(2,16). In the last three cases the full design groups contain the indicated native Mathieu groups, and their orders are 887,040,10,200,960,244,823,040 respectively.

Proof. Suppose first that Γ(S) is firm and G is flag-transitive. The firmness criterion gives $k-t+1\ge 2$ and $r=(v-t+1)/(k-t+1)\ge 3$. The equivariant chamber bijection of Theorem S.3 makes G t-transitive on the points of S. Kantor's Theorem3, retained with its precise primary-source hypotheses, therefore leaves the binary affine plane family, the projective subline family, and the five Mathieu designs.

For the binary affine plane family S_d, the one-point derivative is PG(d−1,2). The residue theorem makes Γ(S_d) residually connected exactly when its rank-three derived construction is connected. B.ProjectivePlane includes the positive case PG(2,2), while B.SpaceObstruction excludes every projective dimension at least three. Thus d=3. The case d=2 does not satisfy the already necessary condition $r\ge 3$. Conversely the explicit affine action in the binary affine argument above is chamber-transitive, and the same residue argument proves residual connectivity when d=3. Preservation of the fourth point of each affine plane proves Aut(S_d)=AGL(d,2), as proved above.

In the projective subline case the parameters are $S(3,q+1,q^e+1)$, with $e\ge 2$ and $q\ge 3$. The derivative at a point is AG(e,q). B.AffinePlane and B.SpaceObstruction, together with the residue theorem, imply residual connectivity exactly when e=2. For this remaining case Theorem I.Subline proves Aut(S)=PΓL(2,q^2). If $q=p^f$, the chamber count is $|PGL(2,q^2)|q$ whereas the full group order is $|PGL(2,q^2)|\cdot 2f$. Flag transitivity therefore requires q to divide 2f.

This divisibility leaves only q=4 in the range $q\ge 3$. If f=1, it is impossible. If $f\ge 3$, then $q\ge 2^f>2f$, also impossible. If f=2, $q=p^2$ divides 4 only when p=2. At q=4 the normalization in that theorem proves transitivity explicitly: the relevant Frobenius action cycles through the four nontrivial direction classes

and the scalar action covers the three eligible points in each class. This supplies sufficiency as well as necessity. Firmness and $(IP)_2$ follow from the uniform Steiner theorems.

For the two small Witt designs the chamber counts from Theorem S.3 are 23,760 and285,120. The literal finite models and full automorphism computations give orders 7,920 and95,040. A transitive group cannot have an orbit larger than itself, so neither design gives a flag-transitive construction.

For each large Witt design r=5. The explicit finite maps identify every (t−2)-fold derivative with PG(2,4): one derivative is verified against coordinates, and native t-transitivity transports it to every other derivative. B.ProjectivePlane and the residue theorem give residual connectivity. The parameters give firmness, and Theorem S.6 gives $(IP)_2$. The already retained Mathieu fiber computation verifies transitivity of a native subgroup on every stage of the chamber fiber; consequently the full automorphism group is also transitive. This proves sufficiency and completes the classification. ■

The source's theorem already states this final list. The contribution here is a deduction that separates the common structural proofs, retained classification inputs, and exact finite fibers, and that continues to prove the correct final list after correcting the source's intermediate residual-connectivity assertions. It is not a new classification theorem.

Two necessary source qualifications

The unrestricted rank-two lemma

The following counterexample corrects the first equivalence of Lemma 3 of [L], printed p.566. The lemma allows an arbitrary thick connected rank-two geometry S. It asserts that connectivity of the proper-triangle neighbor graphs for S and its dual is equivalent to residual connectivity of Construction3. The implication toward Construction3 fails in that generality.

The example below is the classical cyclic 2-(11,5,2) design, given directly by its difference set so that its properties do not depend on recognizing a named biplane. The contribution here is its application to the precise printed claim and the explicit finite proof. It is not presented as a new design or an open-conjecture resolution.

Let the points and line labels both be Z/11Z, put

$$D = \{1, 3, 4, 5, 9\},$$

and let line a have point set D+a. Thus x is incident with a exactly when x−a∈D.

Proposition. This incidence geometry is thick and connected. It and its dual satisfy the source's property (Δ). Nevertheless Γ(S) has no incidence between a flag object and a line object. The residue at the flag object (1,D) consists of four isolated points, so Γ(S) is not residually connected and is not a geometry.

Proof. Every point lies on five lines and every line contains five points. Among the20 differences d−e with d,e∈D and d≠e, each nonzero element of Z/11Z occurs exactly twice. Consequently every pair of distinct points is on exactly two lines, and every pair of distinct lines meets in exactly two points. In particular S is connected.

It suffices first to check (Δ) at0. Every other point is collinear with0. The five lines through0, with0 removed, are

$$\{3, 5, 6, 7\}, \quad \{4, 7, 9, 10\}, \quad \{1, 5, 8, 10\}, \quad \{1, 2, 6, 9\}, \quad \{2, 3, 4, 8\}.$$

Since every point pair is collinear, an edge {x,y} is near0 exactly when no set in this list contains both x and y. In particular the following nine edges are near0:

$$13,\ 14,\ 17,\ 39,\ 3\,10,\ 45,\ 46,\ 72,\ 78.$$

They form a spanning tree on the ten vertices 1,…,10: vertices 3,4,7 join the root 1, and respectively attach 9,10;5,6;2,8. Thus the neighbor graph is connected. Translation by any element preserves the incidence relation and takes0 to the corresponding point, so (Δ) holds at every point.

The design is explicitly self-dual. Map a line label a, viewed as a point of the dual, to the point −a; map a point x, viewed as a line of the dual, to the line label −x. The incidence test remains x−a∈D. This is a type-preserving isomorphism from S* to S, proving (Δ) for the dual as well.

In Construction3 a flag object (p,a) is incident with line b only if (D+a)∩(D+b)={p}. The intersection has size 5 when a=b and size 2 otherwise, so this never happens. For the particular flag (1,0), the remaining point objects are D∖{1}={3,4,5,9}; there are no remaining line objects. Hence its rank-two residue has four isolated vertices and is disconnected. ■

The point-neighbor graphs are in fact the Petersen graph, but that recognition is unnecessary: the displayed spanning tree proves exactly the needed premise. The original incidence geometry has multiple lines through each point pair, so it is not a partial linear space. Thus this counterexample does not refute the restricted principal classification in L092 Theorem1. Proposition B.Necessity establishes that theorem without using the false unrestricted equivalence.

Chambers and the length of an incidence path

The source says "paths of length four." Its incidence construction instead gives configurations with **four vertices and three edges**:

$$P - M - x - L, \qquad P \neq x, \quad M \neq L.$$

Here P,x are points and M,L are lines of the original partial linear space. The corresponding chamber is (P,L,(x,M)). Partial linearity implies P is not on L. Conversely every chamber yields precisely this path. Thus the correct unconditional equivalence is between chamber transitivity and transitivity on these oriented three-edge paths starting at a point. The author manuscript B2 already specifies this configuration correctly. The source's conventional four-edge reading is a stronger, false statement; no assertion about the author's intended convention is needed.

Take the affine plane AG(2,3) and G=ASL(2,3). It is thick and connected, and G is flag-transitive on it. Construction3 gives a firm residually connected geometry, as proved directly by B.AffinePlane. That proof also establishes that G acts regularly on its chambers. Its group order and chamber count are

$$|G| = 3^2|SL(2,3)| = 216.$$

There are 9·4·2·3=216 oriented three-edge paths beginning at a point. Each extends to exactly two four-edge paths, by choosing a point on the final line other than the middle point. Both choices are distinct from the first point because the first point does not lie on the final line. Hence there are 432 genuine four-edge paths. A group of order 216 cannot be transitive on them. This supplies a counterexample to the conventional four-edge conclusion under all of Theorem1's relevant hypotheses.

More generally, if the finite partial linear space has v points, s points on each line and t lines through each point, the chamber bijection gives exactly $vt(s-1)(t-1)$ chambers: choose P, then M through P, then $x \in M \setminus \{P\}$, then $L \neq M$ through x. Partial linearity ensures $P \notin L$. A group transitive on these chambers has their number as an orbit size, so this integer divides its order. This proves the corrected divisibility assertion of Corollary 1 of [L].

The source's own divisibility formula in Corollary1, v t(s−1)(t−1), equals the three-edge count. It supports the precise chamber-equivalent formulation, but does not prove historical intent about terminology.

Attribution and verification

The incidence constructions, target classifications and named classical designs originate in [L] and its cited predecessors. The common residue theorem and chamber bijection organize the deductions; they do not replace the outside classifications of [BDL] and [K]. The cyclic design used above is classical. The source's exceptional-value, path-length and numerical corrections were established and independently reviewed in earlier project work. Their inclusion here supplies one complete edition, without claiming new historical priority or another solution of the unital conjecture.

The finite calculations use the actual native Mathieu actions. Their role is to verify design blocks, complete automorphism groups, explicit derived-plane maps and stabilizer kernels and complements, not to infer an abstract group or a geometry from its order. The full programs are included separately. In particular, the full automorphism calculation uses GAP together with DESIGN and GRAPE; the independently written finite-field validator checks the saved incidence and action witnesses but does not itself prove completeness of the automorphism search. The projective and affine all-parameter proofs, the common residue recursion, the classification deductions and the Hermitian diameter proof do not depend on the successful execution of small examples.

The finite Lean certificates verify the three displayed derived $PG(2,4)$ incidence maps, the literal disconnected residue over $\mathbb{F}_4$ and the cyclic biplane counterexample. The Hermitian certificate verifies the actual plane transport formula and specified unitary moves; the elementary connectivity theorem is a separate formal component. The identification of the finite Witt tables with native group orbits, full automorphism computations, classical classification inputs and the all-rank geometric deduction are not formalized by these certificates. Each declaration and its exact scope are recorded in the accompanying statement map.

The object descriptions and different type-size multisets of the two rank-five Mathieu residues are proved above, as are the actual stabilizer groups. Their identification with the earlier geometries named $\Gamma_{23}(4)$ and $\Gamma_8(4)$ is not proved here. The accepted principal result also does not extend the source's necessary classification to arbitrary rank-two incidence structures without partial linearity; the cyclic counterexample explains why that qualification matters.

References

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[G] *Flag relations and derived incidence geometries*, the common paper of this topic. Its Part II proves Theorems S.1–S.6 and Part III proves Theorems H.1–H.3, with full proofs.

[BDL] F. Buekenhout, P.-O. Dehaye and D. Leemans, *RWPRI and* $(2T)_1$ *flag-transitive linear spaces*, Beiträge zur Algebra und Geometrie 44 (2003), 25–46, Theorem 6.2 on pp.42–43.

[K] W. M. Kantor, *Homogeneous designs and geometric lattices*, Journal of Combinatorial Theory, Series A 38 (1985), 66–74, Theorem 3 and its proof on p.73.

[P] A. Pasini, *Flag-transitive extensions of dual affine planes*, European Journal of Combinatorics 17 (1996), 657–671. The source cites this work for the two historical residue names. Those names are recorded as unproved identifications here, not used as premises.